Smart Talent Selection Engine

1. Project Overview

The **Smart Talent Selection Engine** is an AI-based recruitment system that automates resume screening and candidate ranking. It allows recruiters to upload multiple resumes, compare them with a job description, and generate a ranked list of candidates based on suitability. The goal is to reduce manual effort in hiring, improve accuracy, and enhance traditional Applicant Tracking Systems (ATS) through AI-based decision-making.

2. Problem Statement

In real-world recruitment, companies receive hundreds or thousands of resumes for a single job. Existing ATS systems mainly rely on keyword matching, which causes several issues:

- Relevant candidates may be rejected due to different wording
- Manual screening is time-consuming
- No explanation for selection decisions
- Poor understanding of context beyond keywords

This project addresses these issues using AI-based semantic matching and intelligent scoring.

3. Proposed Solution

The system automates the recruitment pipeline as follows:

1. Upload multiple resumes via web interface
2. Extract text, skills, and experience from resumes
3. Compare resumes with job description using AI-based semantic similarity
4. Compute a weighted score using multiple parameters
5. Rank candidates and display results on dashboard

4. System Architecture

Workflow

Frontend (React)

↓

Resume Upload API (FastAPI)

↓

Text Extraction (PDF/DOCX)

↓

Skill Extraction (AI + Keyword Matching)

↓

Experience Extraction Module

↓

Job Description Input

↓

Semantic Matching (SentenceTransformer)

↓

Scoring Engine

↓

Ranked Output → Dashboard

Components

- **Frontend (React):** UI for uploads and results
- **Backend (FastAPI):** Handles processing and APIs
- **AI Modules:** Perform semantic understanding and matching

5. Technology Stack

Technology	Purpose
React	Frontend UI
FastAPI	Backend APIs
Python	Core logic
pdfplumber / python-docx	Resume extraction
pytesseract	OCR for images
sentence-transformers	AI-based semantic matching
sklearn	Cosine similarity
Recharts	Visualization

6. AI Ranking & Scoring Logic

The system evaluates candidates using a weighted scoring approach based on three factors: Semantic Score (35%), Skill Score (40%), and Experience Score (25%). The Semantic Score is calculated using SentenceTransformer embeddings and cosine similarity to measure contextual relevance between the resume and job description. The Skill Score is computed by matching extracted skills with required skills using the formula (Matched Skills / Total Required Skills) × 100, along with bonus points for critical skills like Python, SQL, and Machine Learning. The Experience Score is based on years of experience, projects, and certifications extracted from the resume. The final score is calculated using a weighted formula combining all three components along with bonus and penalty adjustments.

7. Key Features

- Automated resume parsing
- AI-based semantic understanding
- Intelligent candidate ranking
- Explainable scoring system
- Interactive dashboard

8. Key Challenges

- Handling unstructured and inconsistent resumes
- Extracting accurate experience information
- Improving skill detection beyond keyword matching
- Balancing scoring weights fairly
- Integrating AI models into real-time backend

9. Future Improvements

- Use advanced transformer models (BERT/GPT-based ranking)
- Add database for persistent storage
- Introduce fairness and bias detection
- Build advanced analytics dashboard
- Integrate real-time job portals
- Add feedback loop to improve ranking accuracy